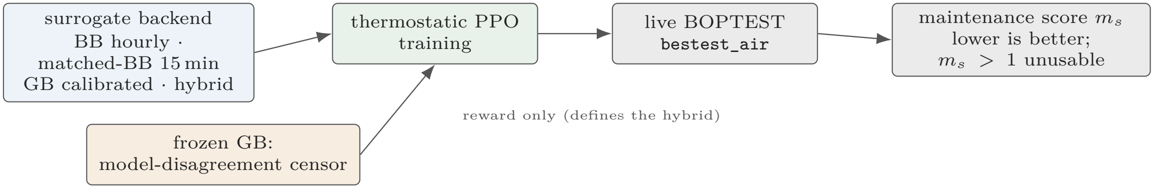
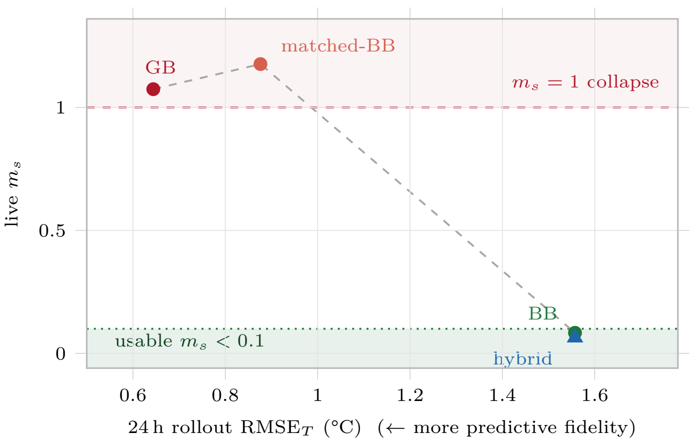
The Fidelity–Utility Paradox in Surrogate-Based RL

Role-Separated Hybrid Methodology for HVAC Control

(a) Framework



(b) Predictive fidelity vs. live utility



Lower rollout RMSE_T does not lower live m_s: the two finer backends cross the collapse threshold, while the hybrid (plotted at BB RMSE_T, since BB rolls out) returns to the usable band.

(c) Role-separated hybrid backend

BB = rollout dynamics (smooth, coarse step)
GB = frozen disagreement censor, reward only
never the rollout model, never the policy loss

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live BOPTEST: $m_s = 0.041$ <5% violation
∼85× throughput

(d) Mechanism

loss of coarse-step smoothing
→ rougher action-response surface
→ bang-bang policy saturation
→ live collapse ($m_s > 1$)

Takeaway: evaluate HVAC surrogates by downstream control utility, not prediction accuracy alone.

Boundary: the censor weight is controller-family specific; the calibration pipeline transfers across three hydronic cases, the frozen policy does not uniformly transfer.